

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
MERN

Practical -2

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Class:TYIT	Batch:1
Date of Assignment:30/06/26	Date/Time of Submission:30/06/26

A).Create Function and call them with Parameters.

Code:-

```
function calculatesalary(name, basicSalary)
{
    let bonus = 5000;
    let totalSalary = basicSalary + bonus;

    console.log("Employee Name:", name);
    console.log("Basic Salary:", basicSalary);
    console.log("Bonus:", bonus);
    console.log("total Salary:", totalSalary);
}
calculatesalary("ashwin",10000);
```

Output:-

```
Employee Name: ashwin
Basic Salary: 10000
Bonus: 5000
total Salary: 15000
```

Code:-

```
function generateBill(product, quantity, price)
{
    let total = quantity * price;
    let bill = total + (total * 0.09);

    console.log("Product:", product);
    console.log("quantity:", quantity);
    console.log("Price Per Item:", price);
    console.log("total Bill:", bill);
}

generateBill("Laptop", 2, 50000);
console.log("ashwin");
```

Output:-

```
Product: Laptop
quantity: 2
Price Per Item: 50000
total Bill: 109000
ashwin
```

B).Work with array and object in javascript.

Code:-

```
let Fruits = ["Mang0", "Banana", "Apple", "Chikoo"];

let student = {
  name: "ashwin",
  age: 20,
  city: "mumbai"
};

console.log("Fruits:", Fruits);
console.log("Second Fruit:", Fruits[1]);
console.log("Student name:", student.name);
console.log("Student Age:", student.age);
console.log("Student City:", student.city);

console.log("ashwin");
```

Output:-

```
Fruits: [ 'Mang0', 'Banana', 'Apple', 'Chikoo' ]
Second Fruit: Banana
Student name: ashwin
Student Age: 20
Student City: mumbai
ashwin
```

C).Demonstrate use of arrow function.

Code:-

```
let add = (a, b) => {  
  return a + b;  
};  
let mul = (a, b) => {  
  return a * b;  
};  
let sub = (a, b) => {  
  return a - b;  
};  
  
console.log("sum = ", add(10, 20));  
console.log("mul = ", mul(10, 20));  
console.log("sub = ", sub(10, 20));  
console.log("Ashwin");
```

Output:-

```
PS C:\Users\s\OneDrive\Desktop\MERN STACK> node prac2c.js  
sum = 30  
mul = 200  
sub = -10  
Ashwin
```

D).Write a program using array methods (map, filter, reduce).

Code:-

```
let numbers = [10, 20, 30, 40, 50];

let doubled = numbers.map(num => num * 2);

let greater25 = numbers.filter(num => num > 25);

let total = numbers.reduce((sum, num) => sum + num,

console.log("Original Array: ", numbers);
console.log("Map:", doubled);
console.log("Filter:", greater25);
console.log("Reduce:", total);
console.log("ashwin");
```

Output:-

```
PS C:\Users\s\OneDrive\Desktop\MERN STACK> node prac2d.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
ashwin
```

E).Create a program that manipulate object and displays outeput.

Code:-

```

let student = {
  rollNo: 101,
  name: "ashwin",
  marks: 85
};

console.log("Original Object:");
console.log(student);

console.log("\nAccessing Properties:");
console.log("Name:", student.name);
console.log("Marks:", student.marks);

student.marks = 90;

console.log("\nAfter Updating Marks:");
console.log(student);

student.city = "Mumbai";

console.log("\nAfter Adding City:");
console.log(student);

delete student.rollNo;

console.log("\nafter Deleting Roll Number:");
console.log(student);

```

Output:-

```

Original Object:
{ rollNo: 101, name: 'ashwin', marks: 85 }

Accessing Properties:
Name: ashwin
Marks: 85

After Updating Marks:
{ rollNo: 101, name: 'ashwin', marks: 90 }

After Adding City:
{ rollNo: 101, name: 'ashwin', marks: 90, city: 'Mumbai' }

after Deleting Roll Number:
{ name: 'ashwin', marks: 90, city: 'Mumbai' }

```

F).Implement a small JavaScript program combining function, arrays and objects.

Code:-

```
let students = [
  {
    rollNo: 101,
    name: "SOM",
    marks: 90
  },
  {
    rollNo: 102,
    name: "Pratik",
    marks: 90
  },
  {
    rollNo: 103,
    name: "Ashwin",
    marks: 90
  }
];

function display(s)
{
  console.log("student Details");
  console.log("-----");
  for(let stud of s )
  {
    console.log(`Roll No : ${stud.rollNo}`);
    console.log(`Name :${stud.name}`);
    console.log(`marks :${stud.marks}`);
    console.log("-----");
  }
}

display(students);
```

Output:-

```
PS C:\Users\s\OneDrive\Desktop\MERN STACK> node prac2f.js
student Details
-----
Roll No : 101
Name :SOM
marks :90
-----
Roll No : 102
Name :Pratik
marks :90
-----
Roll No : 103
Name :Ashwin
marks :90
```